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IMS Nanofabrication GmbH

Recruiting Team

Dresdner Str. 47, 1200 Vienna, Austria

Application for the position: Development Engineer

Dear Recruiting Team,

IMS Nanofabrication drives the limits of lithography at the nanometer scale. As an optical systems engineer finishing my PhD at the Medical University of Innsbruck, I have spent the last four years building complex optomechatronic systems (advanced microscopes) and developing data analysis pipelines to optimize their performance. My background in physics engineering, combined with my exposure to industrial semiconductor workflows at the ASML and ZEISS summer schools, makes me eager to contribute to the development of your Multi Beam Mask Writer systems.

In my current research, I designed and built an automated wavefront sensing and adaptive optics system from scratch. This involved integrating high precision optical components, spatial light modulators, and scientific cameras, while writing the entire control and data acquisition stack in Python and C++. To ensure the system achieved the best potential performance, I developed iterative phase retrieval and optimization algorithms to process wavefront data in real time. This hands-on experience in combining hardware control with advanced signal processing directly aligns with your need for validating new features and analyzing complex log and metrology data.

Beyond system development, I focus on structured testing and quality frameworks. I have structured my instrument development following V-Model systems engineering principles, ensuring clear requirements tracking from initial concept to final validation. By applying Design of Experiments and systematic optical tolerance analysis in python, I have successfully bridged the gap between early stage R&D and stable, reproducible system performance.

I will complete my PhD in June 2026. I am Colombian originally and hold an unrestricted working resident permit in Austria (rot-weiss-rot-plus). I look forward to discussing how my background in optomechanical and electronic systems, data analysis, and structured testing can support your team in Vienna.

Best regards,

Juan David Munoz Bolanos